Approval gates and idempotency

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

One complete approval and idempotency evidence record.

Keep actor, key, payload version, destination, approval scope, stored result, effect count and guarantee limits together.

Keep one habit: Recall → W1–W3/W5 → independent room conflict → project operation ledger → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Planning and action selection. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Logs traces and privacy. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Integration tests and fixtures. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- **Approval gate**s bind side effects to an actor, exact version and destination. Idempotency prevents repeated requests with the same key from repeating a committed effect.

- The mock submission requires approval for request version v1 and destination D. Store key -> committed result. Reusing the same key with the same payload returns that result; the same key with different payload is rejected.

Planning and action selection

> **My remembered idea, example and correction**

Logs traces and privacy

> **My remembered idea, example and correction**

Integration tests and fixtures

> **My remembered idea, example and correction**
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Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main extended mock: actor operator, operation key K, payload send/version v1/destination D/fictional body. Trusted approval A1 binds that actor, key and exact full payload. The first matching call appends one local effect stub and stores copied payload/result. An identical replay returns a copied committed_stub result with one total effect.

  Changing to version v2 with A1 fails approval; a separately matching A2 for v2 still cannot repurpose the old committed K and produces a payload conflict. A1 cannot approve another actor or key K2. Copies isolate caller mutations. Separate desk mock: approved K9 for desk 4, delivered twice, produces one stub; K10 for desk 5 without its own matching approval is denied.

  The original compact approval==payload check is classroom shorthand, not real trusted approval verification. A fresh in-memory ledger forgets the old commit and can append another stub. There is no real actor verification, approval signature, persistence, concurrency safety, external delivery or crash-safe exactly-once guarantee. An uncertain reply alone does not prove the first effect failed.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main extended mock: actor operator, operation key K, payload send/version v1/destination D/fictional body. Trusted approval A1 binds that actor, key and exact full payload. The first matching call appends one local effect stub and stores copied payload/result. An identical replay returns a copied committed_stub result with one total effect.

  Changing to version v2 with A1 fails approval; a separately matching A2 for v2 still cannot repurpose the old committed K and produces a payload conflict. A1 cannot approve another actor or key K2. Copies isolate caller mutations. Separate desk mock: approved K9 for desk 4, delivered twice, produces one stub; K10 for desk 5 without its own matching approval is denied.

  The original compact approval==payload check is classroom shorthand, not real trusted approval verification. A fresh in-memory ledger forgets the old commit and can append another stub. There is no real actor verification, approval signature, persistence, concurrency safety, external delivery or crash-safe exactly-once guarantee. An uncertain reply alone does not prove the first effect failed.

- Approved send v1 to D uses key K, commits once, then caller retries K with the identical payload. Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main extended mock: actor operator, operation key K, payload send/version v1/destination D/fictional body. Trusted approval A1 binds that actor, key and exact full payload. The first matching call appends one local effect stub and stores copied payload/result. An identical replay returns a copied committed_stub result with one total effect.

  Changing to version v2 with A1 fails approval; a separately matching A2 for v2 still cannot repurpose the old committed K and produces a payload conflict. A1 cannot approve another actor or key K2. Copies isolate caller mutations. Separate desk mock: approved K9 for desk 4, delivered twice, produces one stub; K10 for desk 5 without its own matching approval is denied.

  The original compact approval==payload check is classroom shorthand, not real trusted approval verification. A fresh in-memory ledger forgets the old commit and can append another stub. There is no real actor verification, approval signature, persistence, concurrency safety, external delivery or crash-safe exactly-once guarantee. An uncertain reply alone does not prove the first effect failed.

- A report omits this qualification: A key prevents duplicate application effects only within the implemented persistence and downstream guarantees; it is not universal exactly-once delivery.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main extended mock: actor operator, operation key K, payload send/version v1/destination D/fictional body. Trusted approval A1 binds that actor, key and exact full payload. The first matching call appends one local effect stub and stores copied payload/result. An identical replay returns a copied committed_stub result with one total effect.

  Changing to version v2 with A1 fails approval; a separately matching A2 for v2 still cannot repurpose the old committed K and produces a payload conflict. A1 cannot approve another actor or key K2. Copies isolate caller mutations. Separate desk mock: approved K9 for desk 4, delivered twice, produces one stub; K10 for desk 5 without its own matching approval is denied.

  The original compact approval==payload check is classroom shorthand, not real trusted approval verification. A fresh in-memory ledger forgets the old commit and can append another stub. There is no real actor verification, approval signature, persistence, concurrency safety, external delivery or crash-safe exactly-once guarantee. An uncertain reply alone does not prove the first effect failed.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Key J already committed room A. A retry uses key J but room B. What should happen?
- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

Complete task and evidence

- Add a labelled **approval gate**s and idempotency **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **approval gate**s and idempotency.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for **Approval gate**s and idempotency. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Approval gate**s and idempotency. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Key J already committed room A. A retry uses key J but room B. What should happen?

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

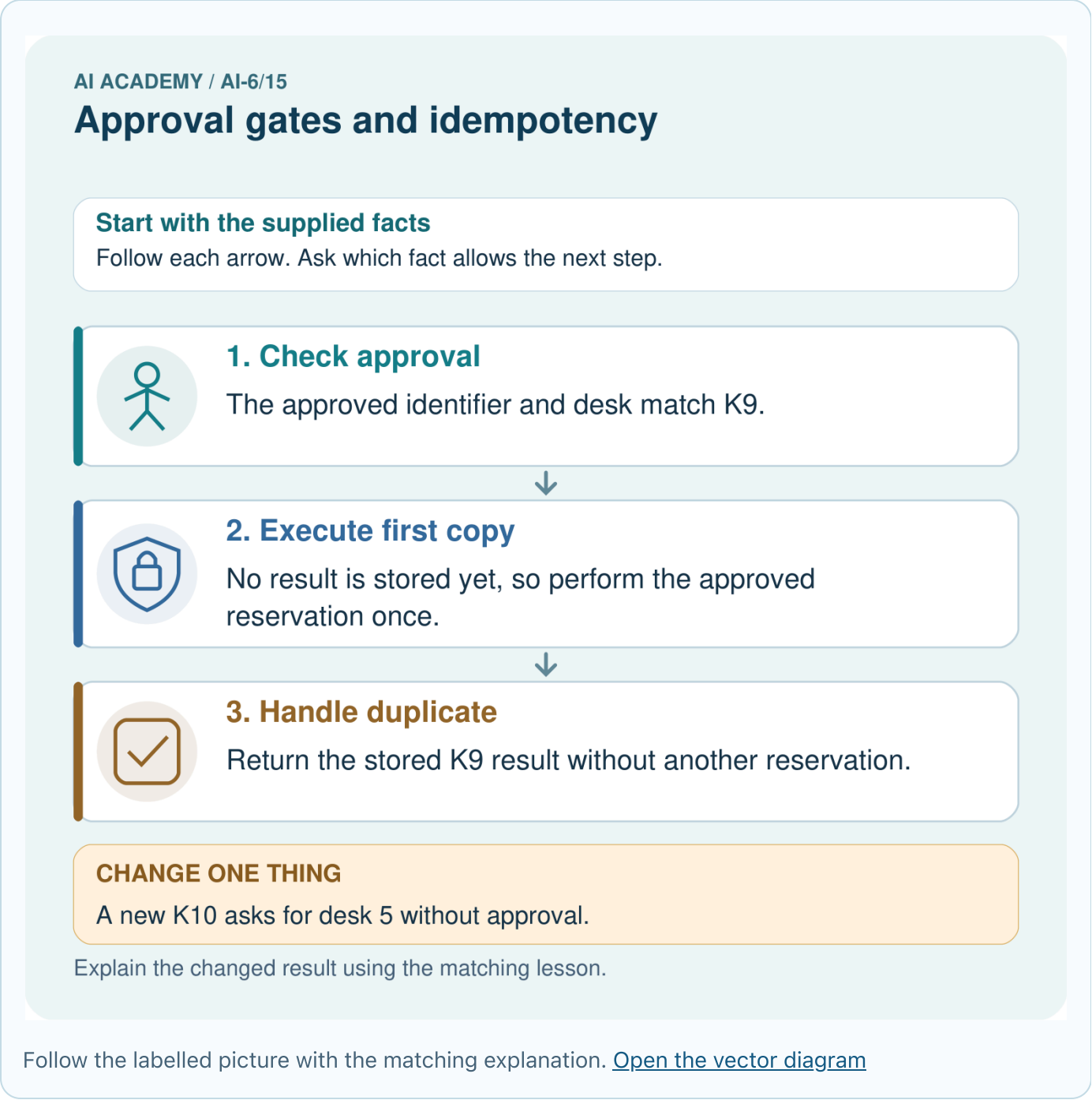
Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- One approval, one reservation. A desk tool needs approval for the exact request and deduplicates repeated request keys.
- K9 reserves desk 4 and is approved once. Two copies of K9 arrive.
- Trace the two deliveries of approved K9. How many reservations should occur, and why can a new K10 for desk 5 not borrow K9's approval?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- One approval, one reservation. A desk tool needs approval for the exact request and deduplicates repeated request keys.
- Which labelled step matches this detail? Return the stored K9 result without another reservation.
- 1. Execute first copy
- 2. Check approval

- 3. Handle duplicate

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check after the first attempt

After your attempt, compare with the supported explanation: Handle duplicate Return the stored K9 result without another reservation. Approval answers whether an action may proceed.

Idempotency prevents repeated handling of that action from creating duplicate effects.

Changed-case question and optional hint

- Changed-case practice: A new K10 asks for desk 5 without approval. Which conclusion is supported by the supplied facts?

- 1. Stop K10 at the **approval gate**.

- 2. We can decide the result without examining the changed **input** or the procedure.

- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Stop K10 at the **approval gate**. K9's approval does not apply to a different action.

Repair and retain the record

Review Planning and action selection (ai-6/14). Goal: Explain the **mechanism** and terminology of planning and action selection. Short practice: Choose a permitted plan when no approved tower-height card exists.

What new answer becomes supported after a card records 24 metres, and why could the bot not invent it earlier?. Return to this week and explain what you changed.

Review Logs traces and privacy (ai-6/12). Goal: Explain the **mechanism** and terminology of logs traces and privacy. Short practice: Identify the purpose of R17, C8 and 240 ms in the trace.

Which field is deliberately omitted, and why would logging the whole question violate this design?. Return to this week and explain what you changed.

Review Integration tests and fixtures (ai-6/9). Goal: Explain the **mechanism** and terminology of integration tests and fixtures. Short practice: Follow request N through the fixture and adapter.

Which source and item should appear in the reply, and what precisely fails when the adapter returns M?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Approval gates and idempotency: the case

AI ACADEMY / AI-6/15 · 01 · LOOK AT THE FACTS

Approval gates and idempotency: the case

The situation

One approval, one reservation. A desk tool needs approval for the exact request and deduplicates repeated request keys.

Supplied evidence

K9 reserves desk 4 and is approved once. Two copies of K9 arrive.

Explain approval gates and idempotency using the deciding step in this fictional service.

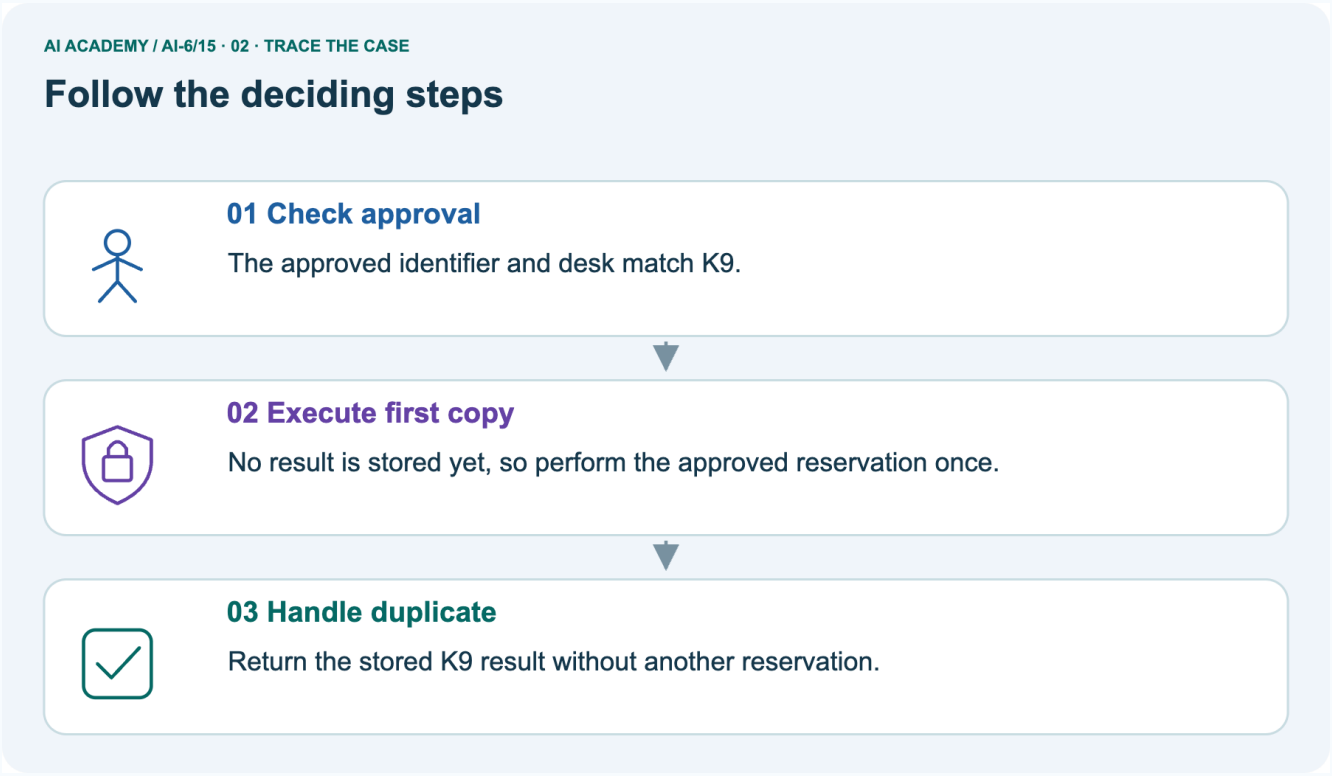
Follow the labelled picture with the matching explanation. Open the vector diagram

One approval, one reservation. A desk tool needs approval for the exact request and deduplicates repeated request keys. Supplied facts: K9 reserves desk 4 and is approved once. Two copies of K9 arrive.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



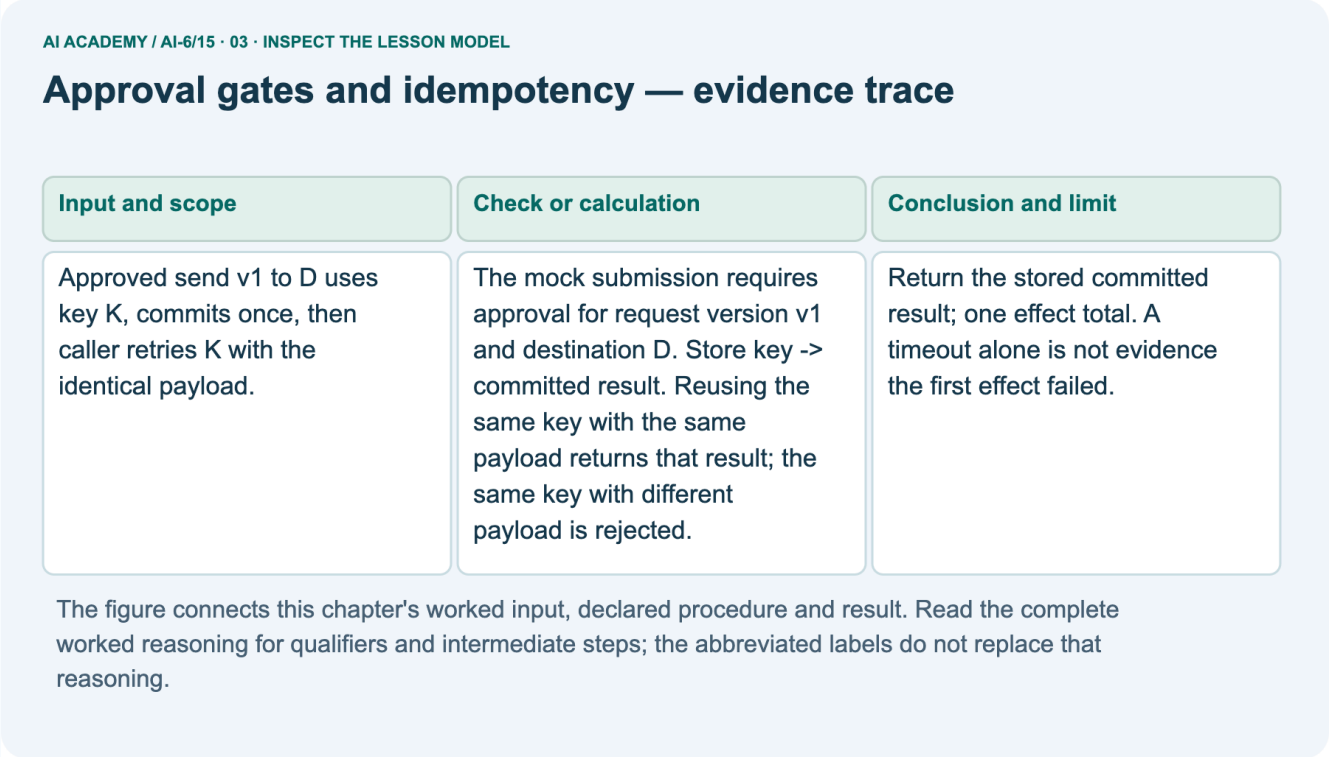
Follow the labelled picture with the matching explanation. Open the vector diagram

Check approval: The approved identifier and desk match K9. → Execute first copy: No result is stored yet, so perform the approved reservation once. → Handle duplicate: Return the stored K9 result without another reservation.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Approval gates and idempotency — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Approval gates and idempotency — **evidence** trace.

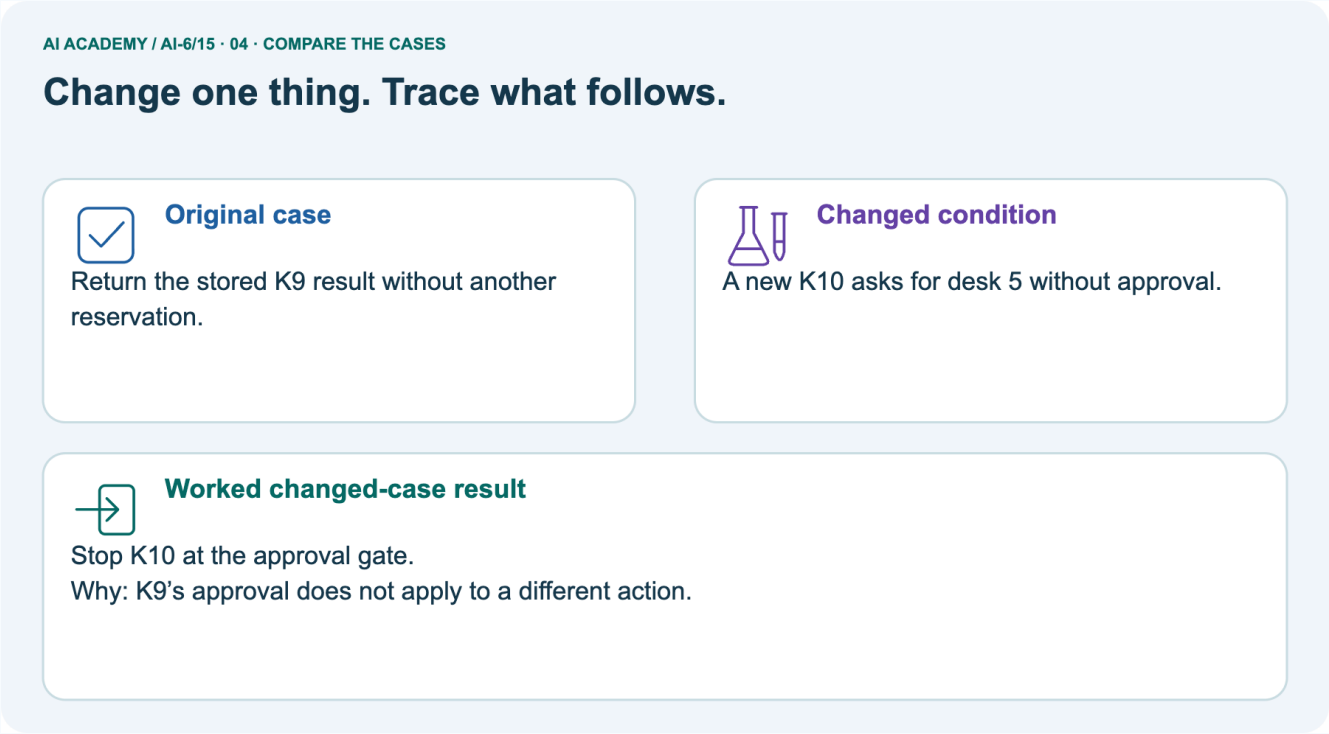
Approved send v1 to D uses key K, commits once, then caller retries K with the identical payload. | The mock submission requires approval for request version v1 and destination D. Store key -> committed result.

Reusing the same key with the same payload returns that result; the same key with different payload is rejected. | Return the stored committed result; one effect total. A timeout alone is not **evidence** the first effect failed.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

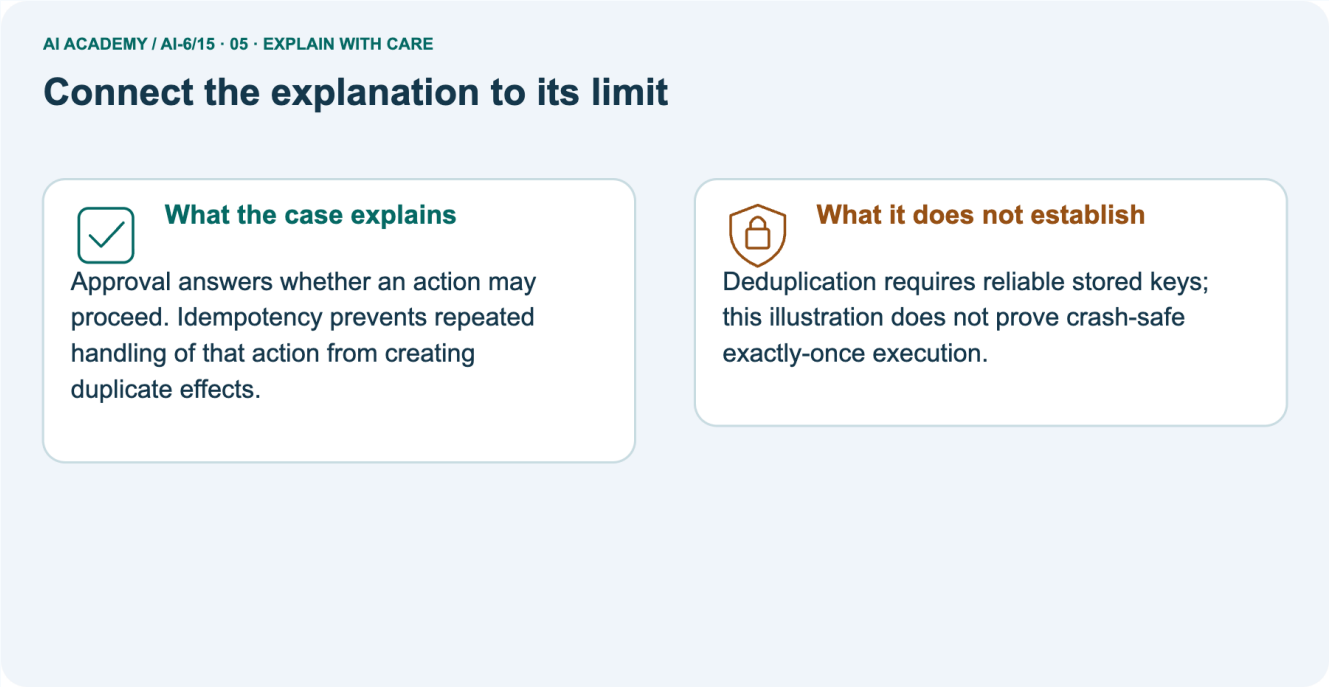
Original case: Return the stored K9 result without another reservation. Changed condition: A new K10 asks for desk 5 without approval. Expected result: Stop K10 at the **approval gate**.

Reason: K9's approval does not apply to a different action.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

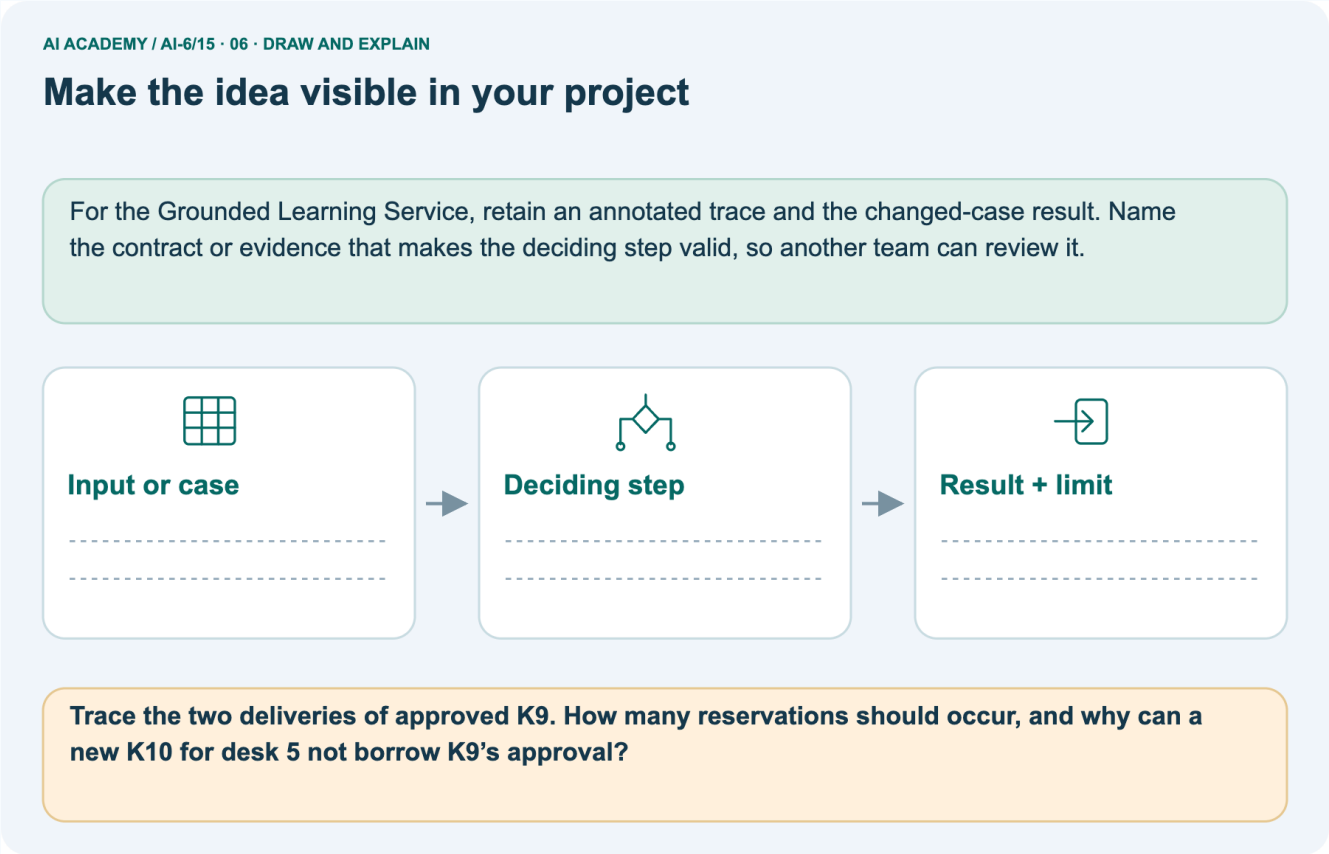
Approval answers whether an action may proceed. Idempotency prevents repeated handling of that action from creating duplicate effects.

Limit: Deduplication requires reliable stored keys; this illustration does not prove crash-safe exactly-once execution.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Trace the two deliveries of approved K9. How many reservations should occur, and why can a new K10 for desk 5 not borrow K9's approval?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Approval gates and idempotency: the case

02. Follow the deciding steps

03. Approval gates and idempotency — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.